
Oversight That Degrades: Architectural Failure Modes in Human-LLM Control Structures

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Abstract

AI safety frameworks widely assume that human oversight provides a reliable control mechanism for AI systems. We present qualitative observational evidence, drawn from longitudinal practitioner contexts, consistent with a recurring degradation pattern under real-world conditions. The failure is not attributable to individual incompetence; oversight architectures appear to degrade structurally. Drawing on observations from academic and enterprise deployments of large language models (2022–2025), we identify a recurring pattern we term *degraded oversight*: human review collapses from active validation into procedural approval under time pressure, scale, and incentive misalignment, effectively transferring decision authority to the AI system while preserving the appearance of human control.

We formalize two distinct control architectures for human-LLM interaction, centaur (gated validation) and cyborg (supervisory control), and illustrate through FMEA-based prioritization that detection difficulty emerges as the dominant concern, outweighing failure severity in the prioritization: the most dangerous failure modes in this analysis are those where oversight has collapsed but the system appears to function normally. These observations are qualitative and practitioner-informed; we present them as a foundation for controlled empirical investigation rather than as generalizable findings in their own right.

1. Introduction

A central assumption in AI safety is that human oversight can constrain AI behavior and catch errors before they prop-

agate into consequential decisions. This assumption underpins regulatory frameworks ([National Institute of Standards and Technology, 2023](#)), deployment guidelines ([Bommasani et al., 2021](#)), and technical safety approaches that treat human review as a meaningful control mechanism. We argue that this assumption may be under strain in ways that existing frameworks do not yet fully specify.

The problem is not that humans cannot oversee AI systems. The problem is that the architectures through which oversight is implemented may degrade under time pressure, scale, repeated exposure, and incentive misalignment. This degradation is not well captured by individual-competence explanations; it appears to be an architectural failure mode, well-documented in decades of human factors research on automation ([Parasuraman & Manzey, 2010](#); [Bainbridge, 1983](#)), but largely absent from current AI safety discourse. Behavioral interventions alone are unlikely to be sufficient when workflow architecture and incentive structures are pushing users toward procedural approval.

We present a control-structure framework for analyzing human-LLM collaboration that makes this degradation visible and tractable. The observations underlying this framework are qualitative and practitioner-informed, drawn from longitudinal involvement in two deployment contexts. We offer them as motivating evidence for the architectural analysis and as a basis for future controlled empirical investigation, not as standalone empirical findings with established generalizability. Adding personnel to a sequentially structured validation chain does not change the architecture; it distributes the same vulnerability across more nodes.

Three observations motivate this work. First, longitudinal observation across academic and enterprise LLM deployments indicates that oversight degradation occurs across contexts, populations, and governance regimes with suggestive consistency. Second, the most dangerous failure modes appear to be precisely those with lowest detectability: systems can appear to function normally while human control has effectively been transferred to the model. Third, behavioral interventions (training, accountability mandates, explainability tools) appear insufficient to prevent degradation because the failure is structural, not behavioral ([Romeo & Conti, 2026](#); [Parasuraman & Riley, 1997](#)).

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This paper contributes: (1) a control-structure taxonomy that maps human-LLM collaboration patterns to safety-relevant failure modes; (2) qualitative observational evidence, consistent across two contexts with opposing governance regimes, that is consistent with an architectural account of oversight degradation and not well captured by individual-competence explanations; and (3) a structured failure mode assessment suggesting that detection difficulty emerges as the dominant concern in the illustrative prioritization, outweighing failure severity across the highest-ranked failure modes.

2. The Oversight Degradation Problem

AI safety research has invested substantially in alignment, interpretability, and evaluation. Comparatively less attention has been paid to a structural question: what happens to human oversight mechanisms under real-world operational conditions?

The human factors literature provides a clear answer. Automation bias, the tendency to defer to automated recommendations even when they conflict with available evidence, is not a bug in human cognition but a predictable attentional phenomenon that affects novice and expert operators alike, resists training interventions, and worsens under multitask load (Parasuraman & Manzey, 2010). Experienced commercial pilots exhibit automation bias under operational conditions, suggesting that expertise does not reliably prevent the effect (Mosier et al., 1998). A systematic review of 35 recent studies confirms that neither accountability mandates nor explanatory interfaces reliably mitigate the effect (Romeo & Conti, 2026).

These findings predate LLMs but transfer directly. The structural conditions that produce automation bias in aviation and process control, high-volume decisions, time pressure, repeated exposure to reliable-seeming outputs, and incentives favoring throughput, characterize LLM deployment in knowledge work. The foundational observation that automating routine tasks paradoxically makes remaining human tasks harder (Bainbridge, 1983) applies with particular force to LLM oversight, where the human’s residual role is precisely the cognitively demanding task of validating fluent, plausible outputs. Lee and See’s trust calibration framework identifies the relevant failure type: degraded oversight represents over-trust miscalibration, in which validators maintain confidence in the oversight process while its functional quality has deteriorated (Lee & See, 2004). Vaccaro and colleagues’ meta-analysis of 106 experimental studies finds that, on average, human-AI combinations performed significantly worse than the best of humans or AI alone under static conditions (Vaccaro et al., 2024), establishing a baseline deficit that the operational degradation trajectory this paper describes compounds.

We term the resulting pattern *degraded oversight*: human review persists in form while losing functional control, such that decision authority transfers to the model despite the nominal presence of a human validator. Our cases indicate this was a recurrent and consequential failure pattern across both observed deployment contexts.

3. Control Structure Framework

We distinguish two architectures for human-LLM collaboration, each with distinct control properties and failure modes. Kasparov’s centaur and cyborg terminology (Kasparov, 2010) has been adopted widely in the human-AI collaboration literature, including by Randazzo and colleagues, whose field study of 244 management consultants identifies centaur, cyborg, and self-automator patterns as empirically distinct collaboration modes (Randazzo et al., 2025). We ground both architectures formally in control theory, treating them as analytical instruments for reasoning about failure modes rather than as exhaustive classifications.

3.1. Centaur Architecture: Gated Validation Control

A centaur architecture implements explicit separation between model computation and human judgment. The LLM generates candidate outputs; humans retain authority over problem framing, validation, and final action through discrete decision gates. Authority concentrates at explicit checkpoints, and transitions from AI output to consequential action are observable and auditable.

The safety-critical property is enforced validation: the system cannot act without affirmative human approval at defined gates. When this property holds, the architecture bounds failure propagation; errors in model output are contained at the validation gate.

The characteristic failure mode is degraded oversight, in which validation gates persist structurally but cease to function as genuine checks. Under time pressure, repeated exposure, and incentive misalignment, human review collapses from critical evaluation into procedural approval. The architecture’s safety properties depend entirely on the quality of human validation; when validation degrades, the architecture silently fails while appearing to operate normally.

3.2. Cyborg Architecture: Supervisory Control

A cyborg architecture integrates LLM capabilities continuously across workflows. Humans set goals, monitor execution, and intervene by exception following the supervisory control paradigm (Sheridan, 1992), rather than approving discrete outputs. Authority distributes dynamically across the human-AI system.

The safety-relevant properties are observability (the human

can detect model state and behavior) and intervention capability (the human can correct or override). When both hold, the architecture enables scalable oversight without per-output validation bottlenecks.

The characteristic failure modes are error propagation (errors cascade through coupled processes before detection), observability gaps (model influence becomes difficult to trace), and responsibility diffusion (unclear authority allocation delays corrective action). These failures differ structurally from centaur failures: they arise from insufficient monitoring rather than degraded gatekeeping.

3.3. Architectural Criteria

Table 1 provides operational criteria for distinguishing the two configurations. Real deployments rarely instantiate either in pure form; these criteria are intended to make the taxonomy inspectable and applicable by other researchers.

Table 1. Operational criteria for architectural classification.

Dimension	Centaur	Cyborg
Authority	Concentrated at checkpoint	Distributed dynamically
Validation	Discrete gate	Continuous monitoring
Auditability	Per-output record	Process-level record
Intervention	Pre-execution approval	Exception-based override
Observability	Output review	State monitoring
Failure sig.	Silent collapse	Diffuse propagation

3.4. Why the Distinction Matters for Safety

Current AI safety discussions frequently invoke “human oversight” or “human-in-the-loop” without specifying the control architecture through which oversight operates. This imprecision obscures a critical safety distinction: centaur and cyborg architectures fail in structurally different ways, require different mitigations, and offer different safety guarantees.

Centaur architectures fail silently; oversight appears intact while control has transferred. Cyborg architectures fail detectably but diffusely; control degradation manifests as increased error rates, slower recovery, and attribution confusion rather than discrete gate failure. Treating both as generic “human oversight” prevents systematic analysis of where and how oversight breaks.

4. Observational Evidence

We draw on qualitative observations from two contexts with opposing governance approaches to LLM deployment, spanning 2022–2025. Observations were gathered through direct practitioner involvement in both settings. Classification of architectural patterns was retrospective, conducted by the author using the criteria in Table 1 applied to assignment artifacts, code review records, retrospective notes, and informal interview records. Assignment artifacts and code review records were retained contemporaneously as part of normal course and workplace practice; retrospective notes and informal interview impressions were not systematically recorded at the time of observation and were reconstructed from memory during analysis. No independent coder was involved; prevalence assessments represent single-observer qualitative judgments, not statistically derived frequencies.

Academic context. A graduate data visualization course across seven semesters (Fall 2022–Spring 2025), approximately 120 students across 14 sections. Standard academic integrity expectations; no formal LLM-specific prohibitions or mandates. Observations from student assignments and analytical artifacts.

Enterprise context. Software engineering and business operations teams (~40 staff) at a mid-size technology company during the same period. An explicit enterprise mandate to “use LLMs in your daily work activity” without prescriptive workflow guidance. Observations from code reviews, documentation, team retrospectives, and informal interviews.

4.1. Convergent Evolution of Oversight Failure

Both contexts exhibited similar three-phase trajectories. We present this as an analytical abstraction imposed on longitudinal observations rather than as an empirically validated stage progression; the phase boundaries are not discrete and the transitions were not measured prospectively.

Phase 1: Isolated usage (2022–early 2023). LLMs treated as curiosities; one-shot queries, simple drafts, localized debugging. Minimal workflow integration, minimal safety concern.

Phase 2: Delegated patterns (mid 2023–early 2024). Systematic delegation with minimal validation. Students incorporated LLM outputs with limited critique; enterprise teams delegated wholesale with involvement limited to prompt formulation and output acceptance. Human review persisted nominally but appeared to provide no functional control.

Phase 3: Architectural differentiation (mid 2024–present). Both contexts diversified into stable patterns. Critically, the distribution was similar despite opposing governance:

Two patterns are analytically relevant. First, degraded over-

Table 2. Architecture distribution across contexts (Fall 2025). Labels represent single-observer qualitative prevalence assessments. Categories are not mutually exclusive architectural states; individual workflows may exhibit multiple patterns across different tasks or time periods.

Pattern	Academic	Enterprise
Atomic (isolated)	Dominant	Dominant
Degraded centaur	Substantial	Substantial
Active centaur	Substantial	Substantial
Experimental cyborg	Small minority	Small minority

sight emerged as a substantial pattern in both contexts despite differing governance stances, a norm-governed non-mandated context in academia and an explicit adoption mandate in enterprise. Second, neither observed site showed evidence that governance stance alone prevented degradation from emerging: neither the norm-governed nor the mandate context appeared to influence how users allocated authority once engaged with LLMs.

4.2. Illustrative Cases

The following cases are presented as concrete instances of the patterns described above, not as representative samples or statistical evidence.

Academic case. In the graduate data visualization course (Spring 2024), a student submitted an analysis of financial time series data that included a fluent, correctly formatted narrative interpreting a trend that did not exist in the underlying dataset. The LLM had extrapolated beyond the data range; the student had reviewed the prose but had not verified it against the chart. The output was stylistically indistinguishable from a careful human-authored analysis. The error was identified only during grading through direct comparison of prose claims against plotted values. This instance is consistent with Phase 2 dynamics: validation persisted in form while ceasing to function as a genuine check.

Enterprise case. Across approximately eight sprint retrospectives at the enterprise site (late 2023 to early 2024), engineering team members reported that pull request review times for LLM-assisted code had decreased substantially relative to human-authored code, while reviewer comment rates had also decreased. Team members attributed both changes to improved output quality. Post-hoc review of a convenience subset of approximately 30 merged pull requests identified instances where logical errors in LLM-generated code had passed review without comment; no systematic comparison against a matched baseline was conducted. The pattern is consistent with output fluency suppressing scrutiny: well-structured outputs were processed as probably correct without the verification that would have

caught the errors. This case is described at the level of pattern and mechanism; the organizational context is not identified.

4.3. Limitations of Observations

These observations derive from two contexts using qualitative, single-observer methods. The classification protocol was retrospective and lacks independent verification. Distributions are approximate. The absence of systematic disconfirming case analysis is a structural limitation of the design; absence of observed counterexamples reflects the scope of the evidence base rather than their non-existence. The convergence across contrasting governance regimes provides some preliminary confidence in the structural account, but does not establish generalizability. These observations are presented as motivating evidence and as a basis for future controlled empirical investigation.

5. Failure Mode Analysis

We apply Failure Mode and Effects Analysis (FMEA) as a structured heuristic for prioritizing safety-relevant failure modes across architectures. FMEA scores each failure mode on Severity (S), Occurrence (O), and Detection difficulty (D), each 1–10, yielding a Risk Priority Number (RPN = $S \times O \times D$). Scores represent the author’s structured single-scorer judgment, informed by the observational evidence and systems engineering practice. The table should be read as illustrative prioritization: it makes the relative weighting of failure modes explicit and inspectable, not as a measurement instrument with independent empirical standing. Sensitivity to scoring variation has not been formally assessed. One-factor ± 1 perturbations of top-ranked scores can alter the exact ordering within the degraded-centaur tier (for example, lowering Authority Drift’s occurrence score by one point moves its RPN to 441, below Blind Acceptance at 448), but both degraded-centaur entries remain higher-ranked than all other failure modes under small perturbations. The stable finding is that the two degraded-centaur modes occupy the top tier as a set; their exact ordering within that tier is sensitive to scoring assumptions.

5.1. Detection Difficulty as Risk Multiplier

The FMEA suggests a pattern with direct safety implications: detection difficulty ($D \geq 8$) characterizes all highest-priority failure modes. The most dangerous failures in human-LLM systems are precisely those where the system appears to function normally.

Authority Drift (RPN 504) receives an occurrence score of $O=8$ because both observed deployment contexts showed this pattern emerging as a stable operational state rather than an episodic failure; its detection score of $D=9$ reflects the

Table 3. FMEA: Safety-relevant failure modes in human-LLM architectures.

Arch.	Failure Mode	S	O	D	RPN
Deg. Centaur	Authority Drift	7	8	9	504
Deg. Centaur	Blind Acceptance	8	7	8	448
Act. Centaur	Deployment Coercion	9	5	9	405
Cyborg	Observability Gaps	6	7	9	378
Deg. Centaur	Validation Erosion	6	8	7	336
Cyborg	Error Propagation	8	6	7	336
Cyborg	Responsibility Diffusion	7	7	6	294
Act. Centaur	Rejection Loop Bypass	7	4	5	140

structural indistinguishability of Phase 2 validation from Phase 1 validation in organizational monitoring records. Authority Drift outranks Deployment Coercion (RPN 405) despite lower severity because its detection score makes it nearly invisible to organizational monitoring. Decision authority transfers gradually; each accepted recommendation establishes precedent, efficiency pressures make validation feel like friction, and boundaries erode imperceptibly.

This suggests a specific implication for AI safety: reactive governance may underperform when the highest-risk failure modes are defined by their undetectability. Systems that rely on detecting oversight failures after they occur may systematically miss precisely those failure modes. Safety mechanisms may need to be embedded proactively in the control architecture rather than layered on as monitoring.

5.2. Degraded Centaur as Primary Risk

Degraded centaur architectures concentrate the two highest-priority risks (RPN 504, 448). The occurrence scores ($O = 7-8$) reflect the author’s judgment that these patterns were recurrent in the observed contexts and should be treated as operationally important rather than exceptional; they are not independent evidence of real-world prevalence rates.

The mechanism is structural. When validation depends on individual discipline under time pressure and efficiency incentives, degradation becomes architecturally predictable. The RPN differential between degraded centaur (504/448) and active centaur (405/140) is consistent with the safety value of enforced versus discretionary validation being substantial.

This analysis challenges a common assumption in AI safety: that human oversight is a binary property (present or absent). Our framework suggests oversight exists on a degradation continuum, and a plausible and operationally important state is partial degradation, present in form but diminished in function.

5.3. Organizational Dynamics Override Architecture

Deployment Coercion (RPN 405) represents a distinct failure class: organizational dynamics overriding technical safety controls. Junior employees face pressure to approve systems with recognized risks, deploying known-deficient systems while bearing accountability without authority.

Mitigation requires governance addressing organizational culture, including multi-party approval, anonymous escalation, and protected disclosure, rather than purely technical controls. Architectural rigor alone is insufficient for safety; organizational context determines whether architectural properties hold in practice.

6. Implications for AI Safety

6.1. Scalable Oversight

The scalable oversight problem is typically framed as a capability challenge: how can humans evaluate outputs they may not fully understand? Our analysis identifies a complementary structural challenge: even when humans *can* evaluate outputs, the oversight architecture appears to degrade under operational conditions until they effectively *don’t*.

This suggests scalable oversight may require not only better evaluation tools but architectural designs that resist degradation. Candidate design responses include enforced validation gates, minimum review time requirements, and periodic architecture audits to address the structural dimension. Whether these mitigations are effective under operational conditions is an empirical question this paper does not answer. The cyborg alternative, supervisory control with strong observability, may scale better than centaur architectures for high-volume decisions, but introduces its own failure modes around error propagation and responsibility diffusion.

6.2. Safety by Design

The FMEA results are consistent with a candidate design principle: safety-critical human-AI systems should be designed for the degraded state, not the ideal state. If degraded oversight is the predictable operational condition, then validation gates may warrant technical enforcement rather than procedural suggestion; detection mechanisms might target the absence of genuine review rather than the mere absence of review; and risk assessment could evaluate architectures under degraded conditions rather than optimal ones. These are design hypotheses that follow from the failure mode analysis; their effectiveness under operational conditions requires empirical investigation.

6.3. Governance and Regulation

Current AI governance frameworks do address oversight. The NIST AI RMF asks organizations to define human roles, document generalizability limitations, and gather evidence about human-AI configurations; the EU AI Act requires effective oversight, correct output interpretation, override authority, and explicit awareness of automation bias (National Institute of Standards and Technology, 2023; European Parliament and Council, 2024). The manuscript’s argument is therefore not that governance ignores oversight, but that existing frameworks do not yet differentiate centaur gate-collapse from cyborg propagation failure with the architectural specificity this paper proposes. Specifying oversight without specifying the control architecture through which it operates and the conditions under which each architecture degrades may leave that oversight less reliable than frameworks assume.

7. Related Work

This work bridges human factors engineering and AI safety. The automation bias literature (Parasuraman & Manzey, 2010; Parasuraman & Riley, 1997; Mosier et al., 1998) documents oversight degradation in traditional automation but has not been systematically applied to LLM deployment. Systems safety research, particularly Leveson’s STAMP framework (Leveson, 2012), provides the theoretical basis for analyzing failures as control structure problems rather than component failures. Woods and Hollnagel’s joint cognitive systems perspective (Woods & Hollnagel, 2006) argues that human-machine performance is an emergent property of system structure, consistent with our finding that oversight quality appears to depend on architecture rather than individual capability.

Within AI safety, the closest related work addresses scalable oversight (Amodei et al., 2016; Shneiderman, 2020), AI governance (Raji et al., 2020; Batool et al., 2025), and the jagged frontier of human-AI collaboration (Dell’Acqua et al., 2023). Randazzo and colleagues’ empirical taxonomy of collaboration modes identifies centaur, cyborg, and self-automator patterns as distinct in practice and consequential for skill development (Randazzo et al., 2025); our contribution is the structural analysis of why those configurations degrade differently under sustained operational pressure. We formalize the control-structure dimension that connects these threads: oversight appears to fail not because the AI is too capable or the human too limited, but because the architecture through which they interact degrades predictably.

8. Limitations and Future Work

Our observations derive from two knowledge-work contexts, a graduate professional course and a mid-size technology

company, using qualitative, single-observer methods across 2022–2025. The same author observed the settings, generated the framework, coded the evidence, and interpreted the results. This paper is best understood as disciplined practitioner theorizing: the observations are consistent with the architectural account and motivate it, but they do not independently validate it. The convergence across contexts with opposing governance regimes provides some preliminary confidence in the structural account but does not establish generalizability. Distributions are single-observer qualitative prevalence assessments, not measurements. The retrospective design and absence of independent coding mean the classification reflects one practitioner’s analytical judgment; the methodology cannot systematically surface disconfirming cases, and their absence in this account reflects the scope of the evidence base rather than their non-existence.

The three-phase degradation pattern is characterized qualitatively. We do not specify thresholds at which transitions become likely: at what output volume, deployment duration, or model count does a centaur architecture move from Phase 1 to Phase 2? The evidence base does not support threshold quantification. Identifying reliable transition indicators and the organizational variables that modulate them is a direct direction for future empirical work. Primary observations are drawn from knowledge-work contexts; domains with different feedback structures, accountability norms, or consequence profiles may exhibit different transition rates or detection signatures.

The supervisory constraint architecture discussed in Section 6 rests on theoretical and analogical grounds. Within our observation window, we are not aware of such configurations deployed at operational scale in LLM-assisted knowledge work; the implications of Section 6 are analytically derived and should be read as structurally motivated proposals warranting empirical investigation, not prescriptive design conclusions.

Earlier practitioner experience across configuration management, access control infrastructure, and algorithmic model governance settings suggested that structurally analogous degradation patterns preceded the LLM era. Those earlier contexts are not catalogued here with the same methodological specificity as the two focal sites; they are background intuition rather than a second systematically described evidence base, and should be read accordingly. The working hypothesis they support is that LLMs did not introduce the failure mode but may have accelerated the structural conditions that produce it.

Key directions for future empirical work: Can degradation be predicted from leading indicators such as declining correction density or decreasing override rates? Which architectural mechanisms most effectively resist degradation under operational pressure? Do these patterns replicate in safety-

critical domains such as healthcare, autonomous systems, or critical infrastructure?

9. Conclusion

Human oversight of AI systems is widely assumed to be a reliable safety mechanism. We present architectural analysis and qualitative observational evidence consistent with the view that this assumption may be under strain: oversight appears to degrade from active validation to procedural approval under operational conditions, and the resulting failure modes are characterized by low detectability. Systems can appear safe while human control has effectively been transferred to the model.

The observations motivating this framework are drawn from two LLM deployment contexts spanning 2022–2025. Earlier practitioner experience across configuration management, access control infrastructure, and algorithmic model governance settings provides background intuition that structurally analogous degradation patterns preceded the LLM era; those earlier contexts are not catalogued here with the same methodological specificity as the focal sites and should be read accordingly. The working hypothesis they support is that LLMs did not introduce the failure mode but may have accelerated the structural conditions that produce it. Organizations currently deploying LLM-assisted workflows may be recreating conditions that produced documented failure in earlier technological contexts. The pattern begins again in Phase 1, which looks like success.

What large language models contribute to this picture is acceleration, not novelty. Fluent outputs reduce scrutiny triggers. Rapid model proliferation creates validation demands that outpace available infrastructure. Governance frameworks written before generative AI was a deployment variable were not designed with the detection difficulty described in Section 5 as a parameter. This is a control structure problem. The analytical vocabulary now exists to describe it precisely. The harder work is designing oversight infrastructure for the operational conditions that actually exist.

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